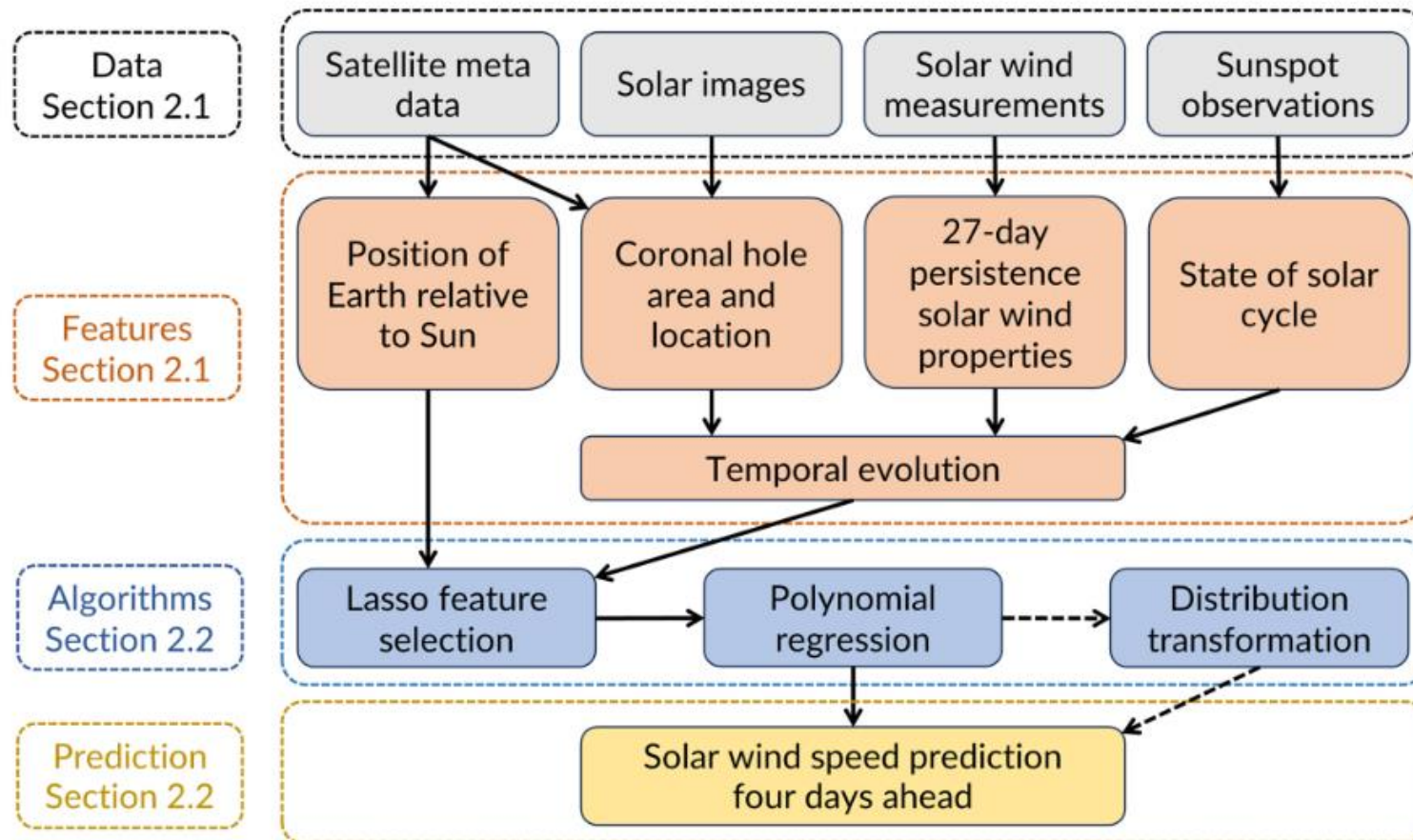


Collin, D., Shprits, Y., Hofmeister, S. J., Bianco, S., & Gallego, G. (2025). Forecasting high-speed solar wind streams from solar images. Space Weather



For coronal hole in each of the 6 grids, compute:

- average intensity
- median intensity
- minimum intensity
- mean of 25% darkest pixel

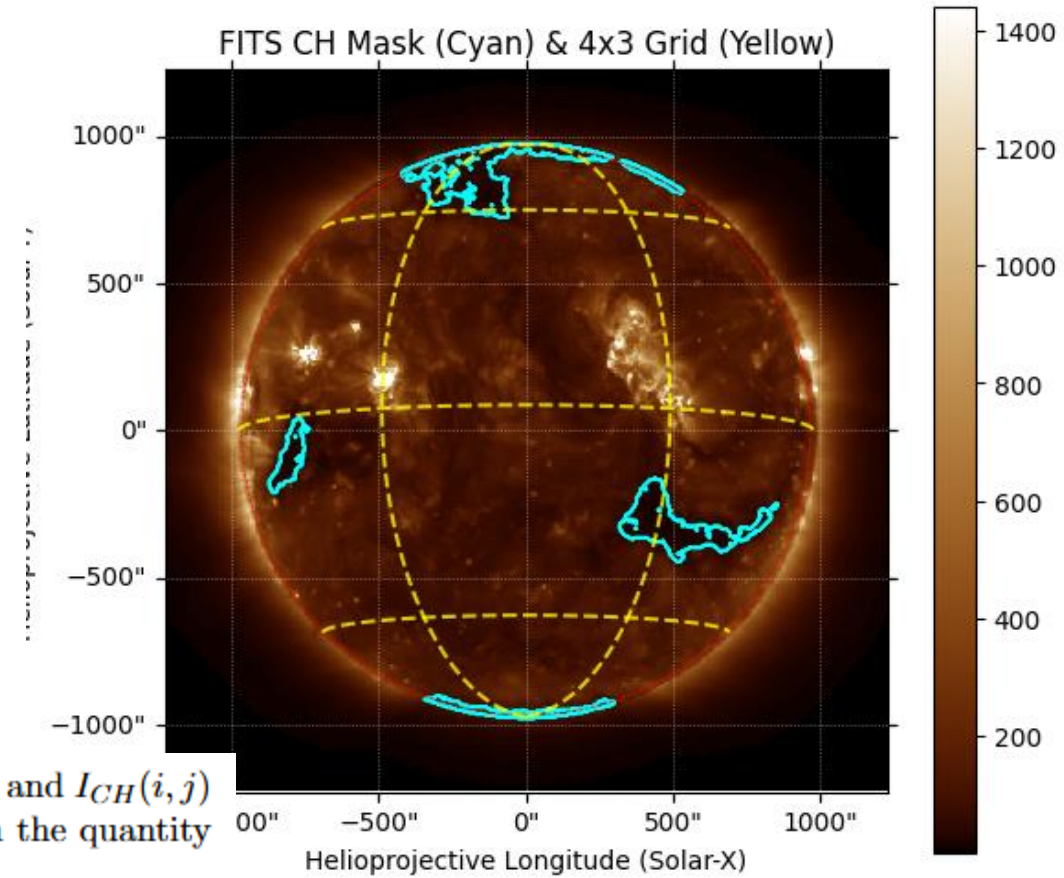
For 193 and 211 channel from 2010 to 2019.

Let I_{QS} denote the median of the solar intensity on the disc ($\sim$ background intensity) and $I_{CH}(i, j)$ a summary statistics (e.g. min, median, etc,...) of CH intensity in the (i, j) cell. Then the quantity we could use to forecast the solar wind speed is something like:

$$\text{proxy}I_{CH}(i, j) = \frac{I_{QS} - I_{CH}(i, j)}{I_{QS}} \quad \text{if } A_{i,j} \neq 0 \quad (1)$$

$$= 0 \quad \text{if } A_{i,j} = 0 \quad (2)$$

~ 100 new features



Correlation between 4-day delayed new_data and Solar Wind Speed (Y):

1 to 25 of 114 entries

Filter

index	Solar_Wind_Speed_Y
global_qs_median_193	-0.0782385413542479
global_ch_mean_193	-0.2250049319301486
global_ch_median_193	-0.2515955365433429
global_ch_min_193	-0.21601410362660464
global_mean_darkest_25_193	-0.24933747417715243
Equatorial_Lon0_ch_mean_193	-0.02900831162746042
Equatorial_Lon0_ch_median_193	-0.04135711771953869
Equatorial_Lon0_ch_min_193	-0.056357848558273506
Equatorial_Lon0_mean_darkest_25_193	-0.05161694554919255
Equatorial_Lon1_ch_mean_193	-0.38863698420295284
Equatorial_Lon1_ch_median_193	-0.3949837630162695
Equatorial_Lon1_ch_min_193	-0.37587126935859155
Equatorial_Lon1_mean_darkest_25_193	-0.3837742300638849
Equatorial_Lon2_ch_mean_193	-0.031094636348802673
Equatorial_Lon2_ch_median_193	-0.036807970485162926
Equatorial_Lon2_ch_min_193	-0.07679527398852001
Equatorial_Lon2_mean_darkest_25_193	-0.06565795092321058
Polar_Lon0_ch_mean_193	-0.03463808486235178
Polar_Lon0_ch_median_193	-0.045790275495679954
Polar_Lon0_ch_min_193	-0.06327807224714493
Polar_Lon0_mean_darkest_25_193	-0.05819508902514498
Polar_Lon1_ch_mean_193	-0.008223060263691923
Polar_Lon1_ch_median_193	-0.02396267579689486
Polar_Lon1_ch_min_193	-0.04244748976772328
Polar_Lon1_mean_darkest_25_193	-0.03431214578937501

Show 25 per page

1

2

3

4

5

Features from the central earth have significantly higher correlations.

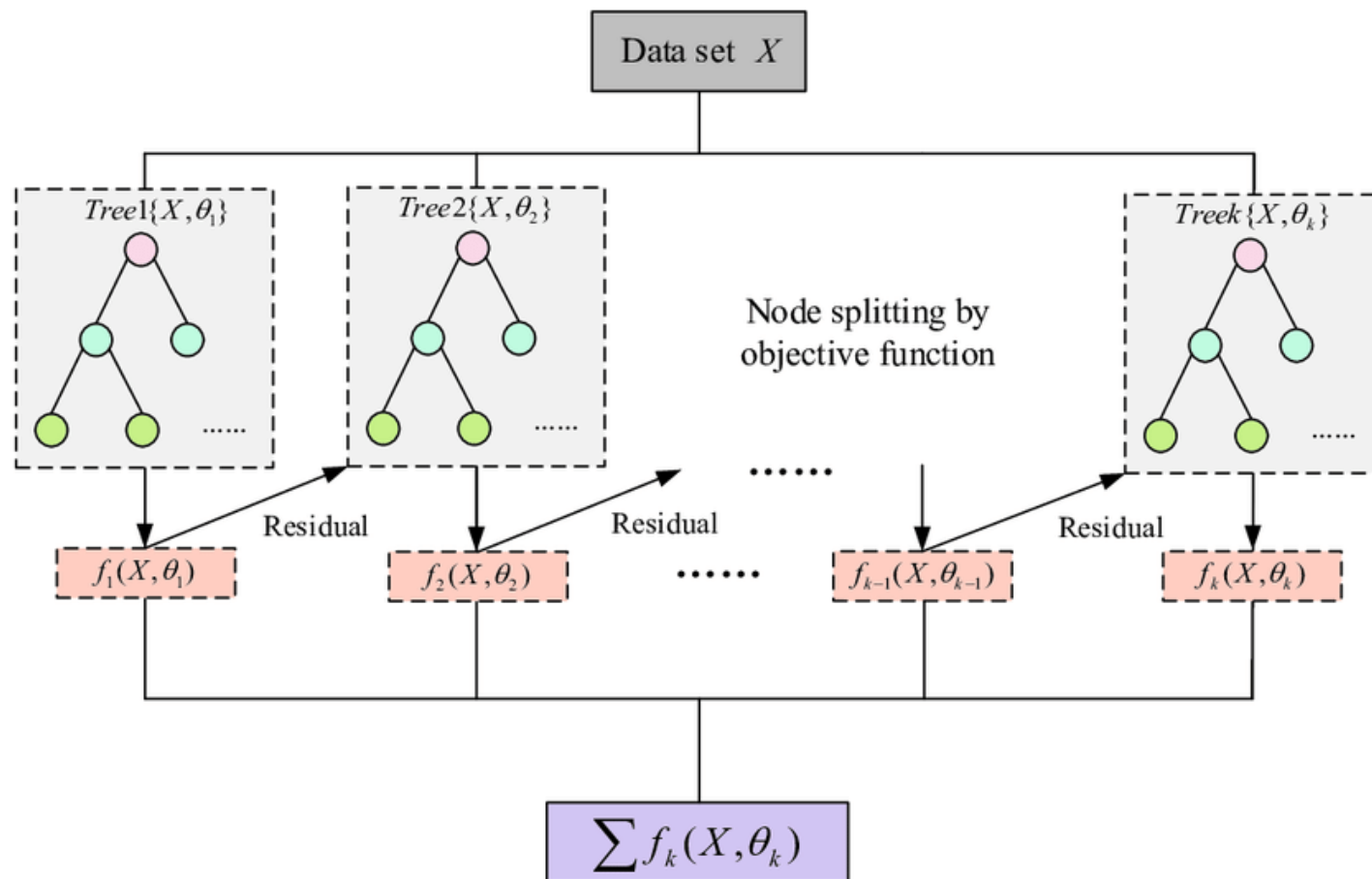
Correlations from 193 are generally higher than those from 211.

Though higher correlations does not guarantee better predicting contributions.

Potential extension: consider using these features from 5/6/7 days ahead.

Model	RMSE	MAE	CC
Constant predictor	97.4	76.2	-0.30
27 persistence	89.1	65.4	0.56
Linear regression	71.2	54.8	0.67
Polynomial(Collin)	68.1	52.6	0.70
Polynomial(us)	68.4	52.6	0.70
Simple MLP	70.7	53.6	0.67
Polynomial with grid intensities	68.2	52.5	0.70
Above + contrast	68.6	52.6	0.69
XGBoost	66.1	51.5	0.63
XGBoost with grid intensities	64.8	50.3	0.64
Above + contrast	64.5	49.9	0.65

XGBoost regression:



Feature importance analysis

- For XGBoost with intensity and contrast features

